

TOMÁS ALADJEM RAMALLO

Junior Computer Engineer | Software · Backend · Full Stack · AI

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LinkedIn · GitHub

PROFILE

Computer Engineering graduate from the University of Barcelona with professional experience in Full Stack and backend development at Win Systems. Experience in REST APIs, client-server architectures, web applications, and server monitoring systems using C#, SQL, and JavaScript. Additional experience in multimodal AI and autonomous multi-agent systems.

PROFESSIONAL EXPERIENCE

Win Systems — Full Stack Software Engineer Intern

Barcelona, Spain · 2025–2026

- Developed a centralized web platform for **server monitoring and log management**, eliminating the need for individual remote access to each machine.
- Designed and implemented **REST APIs in C#** to collect, process, and expose server and infrastructure data.
- Developed a **client-server architecture** for collecting logs and metrics from multiple servers.
- Implemented log processing, including detection of **errors, exceptions, and warnings**, syntax highlighting, and real-time monitoring.
- Developed a dashboard to monitor **CPU, memory, storage, databases, and IIS**, using **SQL and Dapper** for data persistence and access.
- Optimized communication between servers and developed concurrent processes and background tasks for periodic monitoring.

PROJECTS

MINDSCAPE — Multimodal AI Platform

React · FastAPI · Python · Gemini 2.5 Flash · REST API · Chakra UI

- Developed a web application combining **image analysis and conversational interaction** using AI.
- Implemented the frontend with **React** and a REST backend with **FastAPI**, integrating Gemini 2.5 Flash for multimodal processing.
- Managed conversational state through conversation history, dynamic prompts, and structured **JSON** responses.
- Implemented generation and download of **conversation summaries in PDF format**.

Autonomous Software Agents — Multi-Agent Autonomous Systems

JavaScript · Deliveroo.js · LLM · A* · PDDL · BDI

- Developed autonomous agents for navigation, package collection, and delivery in dynamic environments.
- Implemented **BDI architecture**, **A*** search, dynamic obstacles, and route replanning.
- Developed **LLM-based agents** and **PDDL planning with Fast Downward**, including communication and multi-agent coordination mechanisms.

EDUCATION

University of Barcelona — BSc in Computer Engineering

2021–2026 · Barcelona, Spain

University of Trento — Erasmus+

2026 · Trento, Italy

TECHNICAL SKILLS

Languages: C# · Python · Java · C++ · C · JavaScript · SQL

Backend: REST APIs · FastAPI · C#/.NET · Dapper · Client-server architectures

Frontend: React · JavaScript · HTML · CSS · Chakra UI

AI: Gemini API · LLMs · Multimodal AI · Prompt Engineering

Databases: SQL · MySQL · Firebase

Tools: Git · GitHub · Bitbucket · Swagger · VS Code · IntelliJ IDEA

LANGUAGES

Spanish: Native · **Catalan:** Native · **English:** C1 (Cambridge English)